

Automation Test Plan

EDSA Traffic Model: Predictions, Insights, & Classifications

Project Information

Project Name	EDSA Traffic Model: Predictions, Insights, & Classifications
Project Description	This project prepares an EDSA road-traffic accident classification pipeline using MMDA records from 2007 to 2016. It standardizes and validates the source data, engineers model-ready features, and will generate accident-risk predictions once the CBADVAI classifier is finalized.
Dataset	RTA_EDSA_2007-2016.csv - Road Traffic Accident Data of EDSA, Metro Manila (2007-2016), sourced from Mendeley Data.
Team Members	Ed Bennett Borromeo - Project Manager / Scrum Master Rovick Dompur - QA / Tester Matthew Fuentes - Full Stack Developer
GitHub Repository	https://github.com/S180-TK/CBSWEG2

Scope note

Unit tests below cover the CBDATSI data-cleaning and feature-engineering functions currently implemented. Model-specific tests will be added after the CBADVAI classifier and prediction target are finalized.

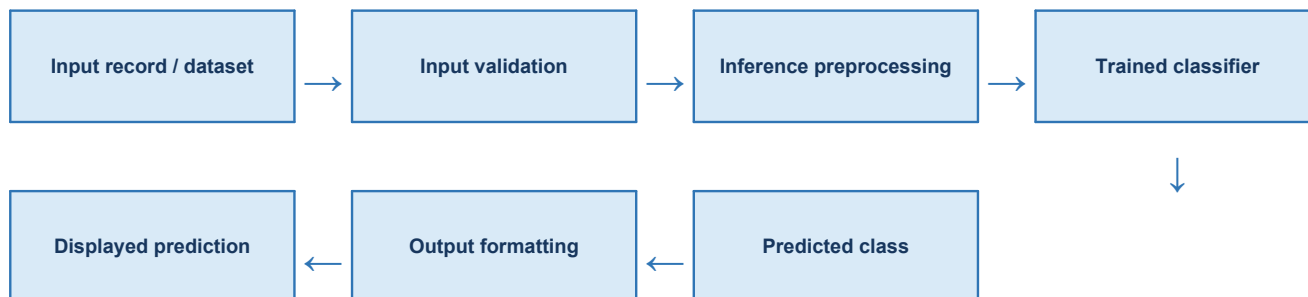
Unit Testing

The following cases correspond to the data-processing functions currently implemented in src/models/data_processing.py.

Module	Data Cleaning and Feature Engineering			
ID	Function	Test Description	Input	Expected Output
EDSA-UT-001	filter_columns()	Keeps only the six required variables and removes unrelated columns.	DataFrame containing the six required columns plus EXTRA_COL	Only ADDRESS, DATETIME_PST, SEVERITY, X, Y, and COLLISION_TYPE remain.
EDSA-UT-002	drop_missing_addresses()	Removes records whose ADDRESS is missing.	['EDSA Guadalupe', None, 'EDSA Cubao']	Two rows remain and ADDRESS contains no null value.
EDSA-UT-003	clean_edsa_address()	Standardizes northbound/southbound tags and barangay abbreviations.	'EDSA N.B. Bgy Guadalupe'	'EDSA (NB) Brgy. Guadalupe'
EDSA-UT-004	clean_edsa_address()	Expands road-type and city abbreviations.	'Ortigas Ave. Q.C.'	'Ortigas Avenue Quezon City.'
EDSA-UT-005	normalize_address_column()	Trims whitespace and applies consistent capitalization before address cleaning.	' edsa guadalupe '	'Edsa Guadalupe'
EDSA-UT-006	extract_hour()	Parses DATETIME_PST, creates HOUR, and removes the original datetime column.	'2016-05-01 14:30:00'	HOUR = 14; DATETIME_PST is removed.
EDSA-UT-007	find_invalid_range_rows()	Returns numeric values outside an accepted inclusive range.	HOUR = [-1, 5, 23, 24]; valid range = 0 to 23	Rows containing -1 and 24 are returned.
EDSA-UT-008	add_severity_num()	Maps severity labels to ordinal values used in the CBDATSI analysis.	['Property', 'Injury', 'Fatal']	[1, 2, 3]
EDSA-UT-009	add_hour_bin()	Groups accident hours into four time-of-day categories.	[0, 7, 13, 19]	['Night (0-6)', 'Morning (6-12)', 'Afternoon (12-18)', 'Evening (18-24)']

System Testing

The final system test will validate the application from input to displayed prediction. The classifier-specific function names will be inserted after the CBADVAI pipeline is finalized.



Excluded from this system flow

Exploratory data analysis, plots, correlation studies, and one-time offline dataset preparation that are not executed during normal inference.

Performance Evaluation

Because the final model is not yet available, targets are stated relative to a baseline rather than as unsupported fixed percentages.

Metric	Formula / Representation	Purpose	Target
Accuracy	Correct predictions / Total predictions	Overall proportion of correct classifications.	Higher than the majority-class baseline.
Precision	TP / (TP + FP), reported per class and as a macro average	Measures the reliability of positive predictions.	Report per class; monitor the minority class.
Recall	TP / (TP + FN), reported per class and as a macro average	Measures how many actual cases in each class are detected.	Prioritize recall for the casualty/minority class.
F1-score	2 x (Precision x Recall) / (Precision + Recall)	Balances precision and recall under class imbalance.	Macro F1 higher than the baseline model.
Confusion Matrix	Actual class by predicted class count matrix	Shows which classes the model confuses.	The casualty/minority class must not be consistently missed.